

Chapter 31: Prior Cesarean Delivery — Revision Notes

Williams Obstetrics 26e, pp. 1475–1511 of the PDF. High-yield numbers are in **bold**. The six tables (31-1 to 31-6) are transcribed from the PDF images (they are missing from the extracted chapter text). All 6 figures are included.

Big picture: "Once a cesarean, always a cesarean" (Cragin, 1916) is still debated 100 years on. **TOLAC** has a small absolute risk of **uterine rupture (~0.5–0.9%)** and slightly higher perinatal mortality/HIE than **ERCD**, but avoids the escalating risks of **multiple repeat cesareans** (previa, **placenta accreta spectrum**, hysterectomy). Most women with **one prior low transverse** incision are candidates.

1. 100 Years of Controversy

- **Eastman (1950):** 30% postcesarean vaginal delivery; **2% rupture**, with **10% maternal mortality** in those ruptures.
- 1960s: cesarean rate only **~5%**; VBAC judged reasonable.
- **1981 NIH conference** + ACOG support (1988, 1994) → **VBAC rate 3.4% (1980)** → **peak 28.3% (1996)**, with a fall in the total cesarean rate.
- Reports of rupture-related morbidity/mortality (McMahon 1996) → **ACOG 1998:** physicians "**readily** available" → **1999:** "**immediately** available". This one-word change is thought to have driven the decade-long **VBAC decline**.
- **TOLAC attempts:** peaked **1995** (just over half), nadir **16% (2006)**, **21% by 2009**.
- **TOLAC success (VBAC among TOLACs):** peak **70% (2000)** → nadir **38% (2006)**.
- 2013: only **1 in 5** with one prior cesarean attempt TOLAC; **<1 in 10** with ≥2 prior (half succeed).
- **Parkland:** only **15%** choose TOLAC, but **86%** succeed.
- **2010 NIH panel** → slight VBAC rise afterwards.

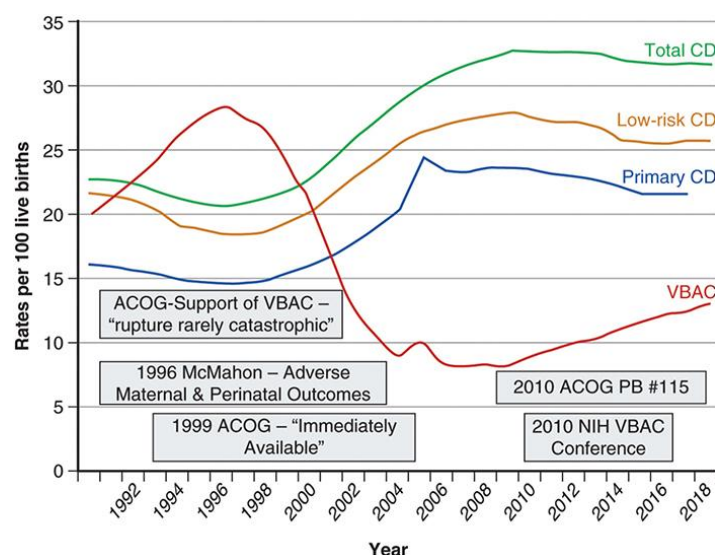


Figure 31-1. US cesarean and VBAC rates 1989–2018: VBAC peaked in 1996, fell after the 1999 "immediately available" wording, and rose slightly after the 2010 NIH conference.

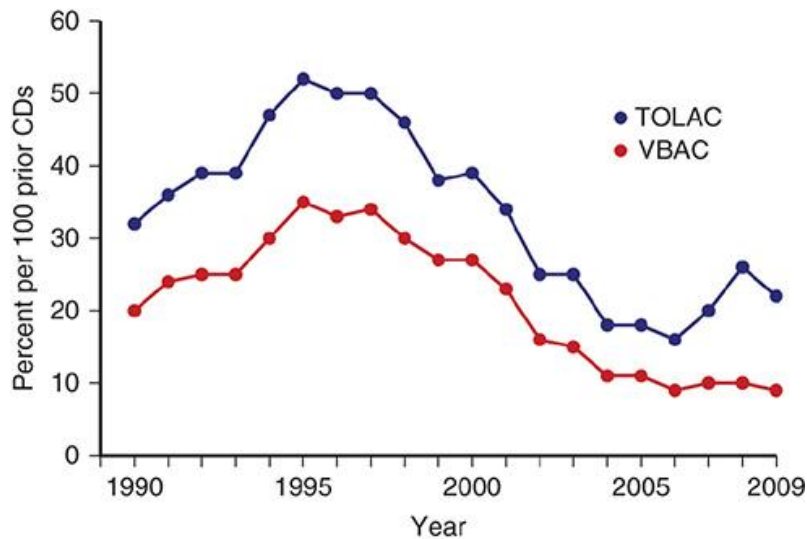


Figure 31-2. TOLAC attempts (peak ~52 per 100 prior CDs in 1995) and VBAC rates (per 100 prior CDs), 1990–2009.

Table 31-1. Acronyms Used to Describe Management of Women with a Prior Cesarean Delivery

Acronym / term	Meaning
CD	Cesarean delivery
TOL	Trial of labor
VBAC	Vaginal birth after CD
TOLAC	Trial of labor after CD
– Failed TOLAC	Resulting in CD
– Successful TOLAC	Resulting in VBAC
ERCD	Elective repeat cesarean delivery
– Scheduled ERCD	CD planned before labor
– Unscheduled ERCD	Planned scheduled CD but indications for early delivery, e.g., spontaneous labor, abruption, etc.
Uterine rupture	
– Complete	All uterine wall layers separated
– Incomplete	Uterine dehiscence with uterine muscle separated but peritoneum intact

2. Influencing Factors

- Counsel **preconceptionally** and early in prenatal care; plans can change as pregnancy evolves.
- **Two choices: TOLAC** (goal VBAC; cesarean during the trial = failed TOLAC) or **ERCD** (scheduled, or unscheduled but planned for labor/other indication).
- Institutional economics, staffing and **medicolegal** factors also shape TOLAC availability.

Table 31-2. Some Factors That Influence a Successful Trial of Labor in a Woman with Prior Cesarean Delivery (CD)

Low-Risk	Favors Success	Increased Failure Rate	High-Risk ^a
Transverse incision	Teaching hospital	Single mother	Classical or T incision
Prior vaginal delivery	White race	Advanced maternal age	Prior uterine rupture
Appropriate counseling	Spontaneous labor	Macrosomic fetus	Patient refusal
Sufficient personnel and equipment	Prior indication of fetal malpresentation	Obesity	Transfundal surgery
	1 or 2 prior low transverse incisions	Current breech	Obstetrical contraindication, e.g., previa
	Nonrecurrent indication	Multifetal pregnancy	Prior open maternal–fetal surgery
	Current preterm pregnancy	Preeclampsia	Inadequate obstetrical or anesthesia staff coverage
		EGA >40 weeks	
		Low vertical incision	
		Unknown incision	
		Labor induction	
		Medical disease	
		Multiple prior CDs	
		Education <12 years	
		Short interdelivery interval	
		Liability concerns	

^aMost consider these absolute contraindications. EGA = estimated gestational age.

3. Delivery Route Risks

Maternal risks

- **Guise review:** TOLAC rupture **absolute risk 0.47%, RR 20.7** vs ERCD.
- **MFMU (Landon 2004):** rupture **0.7%** with TOLAC vs **0** with ERCD. Maternal mortality: no difference or higher with ERCD in various reports.

Table 31-3. Complications in Women with a Prior Cesarean Delivery Enrolled in the Study by the NICHD MFMU, 1999–2002

Complication	TOLAC n = 17,898 No. (%)	ERCD n = 15,801 No. (%)	Odds Ratio (95% CI)	p value
Maternal				
Uterine rupture	124 (0.7)	0	NA	<.001
Uterine dehiscence	119 (0.7)	76 (0.5)	1.38 (1.04–1.85)	.03
Hysterectomy	41 (0.2)	47 (0.3)	0.77 (0.51–1.17)	.22
Thromboembolic disease	7 (0.04)	10 (0.1)	0.62 (0.24–1.62)	.32
Transfusion	304 (1.7)	158 (1.0)	1.71 (1.41–2.08)	<.001
Uterine infection	517 (2.9)	285 (1.8)	1.62 (1.40–1.87)	<.001
Death	3 (0.02)	7 (0.04)	0.38 (0.10–1.46)	.21
Perinatal				
Antepartum stillbirth ^a 37–38 weeks	18 (0.4)	8 (0.1)	2.93 (1.27–6.75)	.008
Antepartum stillbirth ^a ≥39 weeks	16 (0.2)	5 (0.1)	2.70 (0.99–7.38)	.07
Intrapartum stillbirth ^a	2	0	NA	NS
Term HIE^a	12 (0.08)	0	NA	<.001
Term neonatal death ^a	13 (0.08)	7 (0.05)	1.82 (0.73–4.57)	.19

^aDenominator is 15,338 for the TOLAC group and 15,014 for the ERCD group. CD = cesarean delivery; CI = confidence interval; ERCD = elective repeat cesarean delivery; HIE = hypoxic ischemic encephalopathy; NA = not applicable; MFMU = Maternal–Fetal Medicine Units Network; NICHD = National Institute of Child Health and Human Development; NS = not significant; TOLAC = trial of labor after cesarean delivery.

- **Morbidity data conflict:**

- Guise: no difference in hysterectomy or transfusion.
- Mozurkewich meta-analysis: TOLAC **~half** the transfusion/hysterectomy risk.
- MFMU: TOLAC **↑ transfusion and infection**.
- **Canadian cohort (Young 2018):** TOLAC aRR **rupture 6.41, severe PPH with transfusion 2.80**, composite maternal morbidity **1.96**.

- **⚠ Failed TOLAC carries ~5× the major complications of a successful TOLAC.**

Perinatal risks

Outcome	TOLAC vs ERCD
Perinatal mortality (Guise)	0.13% vs 0.05%
Perinatal death (Smith, ~25,000)	1.3/1000, 11× ERCD
Canada: composite severe neonatal morbidity/mortality	aOR 1.49
Canada: neonatal death (failed VBAC / attempted TOLAC)	aOR/aRR 3.22
HIE at term (MFMU)	46 per 100,000 vs 0
Transient tachypnea	3.6% vs 4.2% (↑ with ERCD)
Bag-and-mask ventilation	5.4% vs 2.5%
5-min Apgar, NICU admission	No difference

4. Trial of Labor Selection Considerations

- TOLAC/VBAC **calculators** help predict VBAC success but are **not validated for uterine rupture**; nomograms have poor prognostic value.
- **ACOG**: most women with **one prior low transverse** hysterotomy are candidates → counsel on both TOLAC and ERCD. **Home birth is a contraindication to TOLAC.**

Prior uterine incision

Table 31-4. Types of Prior Uterine Incisions and Estimated Risks for Uterine Rupture

Prior Incision	Estimated Rupture Rate (%)
One low transverse	0.2–0.9
Multiple low transverse	0.9–1.8
Low-vertical ^a	1–7
Classical	2–9
T-shaped	4–9
Prior preterm CD	"Increased"
Prior uterine rupture — Lower segment	2–6
Prior uterine rupture — Upper uterus	9–32

^aSee text for definition. CD = cesarean delivery. Data from the American College of Obstetricians and Gynecologists, 2017; Cahill, 2010b; Chauhan, 2002; Landon, 2006; Macones, 2005a,b; Martin, 1997; Miller, 1994; Sciscione, 2008; Society for Maternal-Fetal Medicine, 2012; Tahseen, 2010.

Incision type

- **One low transverse**: lowest risk of symptomatic separation; asymptomatic dehiscence/"uterine window" still possible.
- **Low vertical without fundal extension**: probably not higher risk than low transverse → **may be a TOLAC candidate** (ACOG).
- **Classical or T-shaped** → **contraindication to labor**. Highest risk: vertical incisions **into the fundus**.
 - Classical scars can rupture **before labor and before term**. Chauhan (157 women, delivery at mean **35.6 wk**): one rupture at **29 wk** (fetal death), **9% dehiscence**, **25% maternal morbidity**; rupture is **neither predictable nor preventable**.
 - Even higher early risk after **open maternal-fetal surgery**.
 - **Prior classical** → **deliver at 36⁰/₇–37⁰/₇ wk** (ACOG).
- **Who gets a classical incision?** 53% of cesareans at **24⁰/₇–25⁶/₇ wk**, **35%** by 28 wk, **<10%** by 32 wk; also noncephalic presentations (a "low vertical" in a preterm breech with an undeveloped lower segment almost always extends into the active segment).
 - Prior **preterm classical**: **~1%** rupture.
 - Prior **perivable** cesarean: rupture **1.8% vs 0.4%** after a term cesarean (half of the perivable ruptures had "low transverse" incisions). Not confirmed by all.
- **Unknown incision**: a potential TOLAC candidate unless the circumstances suggest a classical incision.
- **Müllerian anomalies**: earlier reports suggested ↑ risk; a study of 103 women had **no ruptures**.

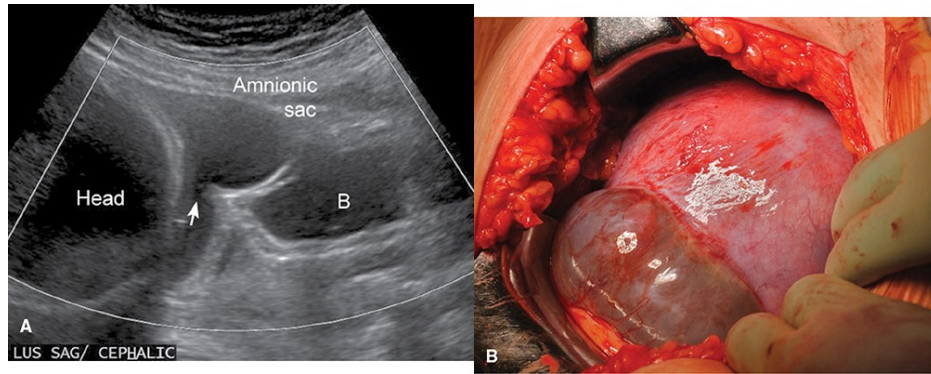


Figure 31-3. "Uterine window" after one low transverse cesarean: sonographic defect with a bulging amnionic sac (A) and the sac seen through the thin lower segment at surgery (B).

Closure technique

- Single vs double layer, locking vs nonlocking: **no difference in dehiscence/rupture** (meta-analysis). Single-layer and locked first layer → thinner residual myometrium on later sonography (another study: double layer didn't thicken it).
- **Parkland**: one running, locking layer.

Number of prior incisions

- Three studies: **2–3× rupture risk** with two vs one prior transverse hysterotomy.
- MFMU: **0.9%** (≥ 2 prior, $n = 975$) vs **0.7%** (one prior), **not significant**.

Imaging the scar

- Large nonpregnant defects and a **thinner lower uterine segment** predict rupture/defect, but **no safe threshold** exists.
- **Jastrow 2016** (LUS measured at **34⁰/₇–38⁶/₇ wk**, $n = 1856$):

LUS thickness	Risk category	TOLAC rate
<2.0 mm	High	9%
2.0–2.4 mm	Intermediate	42%
≥ 2.5 mm	Low	61%

- No symptomatic ruptures in 984 TOLACs. **Parkland does not measure the LUS.**

Prior uterine rupture

- Recurrence: **lower segment up to 6%, upper segment 9–32%**; one series **~20%**.
- ERCD before labor (or at onset of preterm labor) → no ruptures in 60 pregnancies.
- **Deliver at 36⁰/₇–37⁰/₇ wk** (ACOG).

Interdelivery interval

- MRI: full myometrial healing may need **≥ 6 months**.
- **Interdelivery interval ≤ 18 months** → **3× symptomatic rupture** (Shipp; overall rupture 1.4%).
- **Interpregnancy interval <6 months** → **3× rupture**.

Prior vaginal delivery

- Before or after the cesarean → **↑ VBAC success** and **↓ rupture** and other morbidity.

Prior cesarean indication

Prior indication	VBAC rate
Nonrecurring (e.g., breech)	~90%
Fetal compromise	~80%
Labor arrest	~60%

- Prior **second-stage** cesarean → may be linked to second-stage rupture next time.
- Second stage **>3 h** during TOLAC often still ends in VBAC, but rupture/dehiscence rises as it lengthens.

Fetal size and lie

- ↑ fetal size → ↓ **VBAC**; rupture link weaker. Zelop: rupture **1.0% (<4000 g)**, **1.6% (>4000 g)**, **2.4% (>4250 g)** (not significant).
- **Preterm TOLAC**: similar or higher VBAC, **lower rupture**.
- Breech with prior cesarean: little data. **ECV is not contraindicated** by a prior incision (ACOG); success and adverse events similar. (Parkland: no ECV or vaginal breech in these women.)

Multifetal gestation

- Twins do **not ↑ rupture**: rupture **0.7–1.1%**, VBAC **45%** (one study) to **75–85%**.
- **ACOG: twins + one prior low transverse → TOLAC is safe.**

Maternal obesity (VBAC success by BMI)

BMI	VBAC
Normal	85%
25–30	78%
30–40	70%
≥40	61%

Fetal death

- Most prefer vaginal delivery; **maternal risk ↑**.
- MFMU (209 fetal deaths, mean **32.8 wk**): VBAC **87%**; rupture **2.4%** overall, **3.4% with induction**.
- Induction for **second-trimester** demise has a low complication rate.

5. Labor and Delivery Considerations

Timing

- **No nonindicated delivery before 39 completed weeks** (ACOG): ERCD before 39 wk ↑ neonatal morbidity (RDS/TTN, sepsis, any adverse outcome). **Beyond 40 wk** morbidity rises again.
- **Accurate dating** is essential; **early embryo-fetal sonography** is the best method.

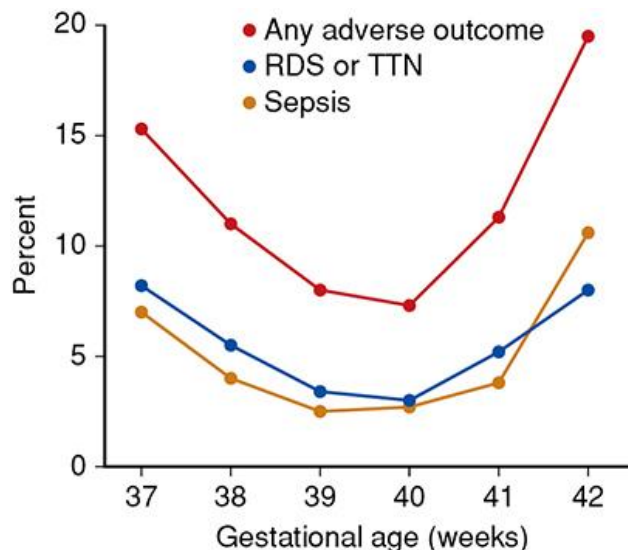


Figure 31-4. Neonatal morbidity with 13,258 ERCDs: lowest at 39–40 wk, higher at 37–38 wk and ≥ 41 wk.

Intrapartum care

- **ACOG:** TOLAC in facilities able to perform **emergency cesarean** for immediate threats to mother or fetus. If unavailable, discuss hospital resources; **referral** may be best.
- Access problem: Ohio hospitals meeting requirements: **15% Level I, 63% Level II, 100% Level III**. TOLAC allowed in **88%** (≥ 1500 deliveries/yr), **59%** (500–1499), **43%** (< 500) of hospitals.

Cervical ripening and labor stimulation

- **Induction** $\rightarrow$ $\uparrow$ **TOLAC failure** vs spontaneous labor.
- **⚠ Misoprostol (PGE₁) is contraindicated** at term with a prior cesarean or major uterine surgery (RCT stopped after **2 of 17** ruptured).
- Ideally avoid induction/augmentation with **unknown incision, unfavorable cervix, or > 40 wk. Parkland: amniotomy only**, no pharmacologic induction/augmentation.

Method	Uterine rupture data
Spontaneous labor (MFMU)	0.4%
Oxytocin induction (MFMU)	1.1%; 1.8% (4x) if no prior vaginal delivery; dose-related
Oxytocin augmentation	0.9%
PGE ₂ gel (Ravasia)	2.9% vs 0.9% spontaneous
Any prostaglandin + oxytocin (MFMU)	1.4% ; prostaglandin alone (n = 227): 0
Sequential PG $\rightarrow$ oxytocin	3x spontaneous
Transcervical Foley (Bujold)	1.6% vs 1.1% spontaneous vs 1.2% amniotomy $\pm$ oxytocin (NS)
Balloon catheter (Sarreau, 101)	50% VBAC, no ruptures
Foley ripening (Hoffman)	6.5% vs 1.9% spontaneous

Epidural analgesia

- Does **not mask rupture**: $< 10\%$ of scar separations cause pain and bleeding; **FHR decelerations are the most likely sign**.
- More frequent epidural dosing was associated with $\uparrow$ rupture.
- VBAC rates similar or higher with epidural; $\sim 1/4$ of VBACs are operative vaginal.
- **AAP/ACOG: epidural may be used safely in TOLAC.**

Uterine scar exploration after VBAC

- Routine transcervical palpation of the lower segment is debated: dehiscence/rupture **0.23%** (3469 VBACs) → explore only with **severe third-stage bleeding** (Silberstein).
- **Parkland** examines routinely. **Laparotomy/repair** for a large tear, **peritoneal breach**, or **active bleeding**.

6. Uterine Scar Rupture

Diagnosis

- Labor progress is similar to normal; **no labor pattern predicts rupture**.
- Early signs can be bizarre: hemoperitoneum → diaphragmatic irritation → **chest pain** (may mimic PE or AFE).
- **Most common sign: nonreassuring FHR** (variable → late decelerations → **bradycardia**). Utah (36 TOLAC ruptures): fetal signs 24, maternal 8, both 3.
- **Contractions rarely stop; IUPC is not reliable** for diagnosis.
- May resemble **abruption** with little pain/tenderness (often masked by analgesia). Diagnosis emerges from **fetal distress ± maternal hypovolemia** (concealed hemorrhage).
- Clues: **loss of station**; presenting part moved away from the inlet; firm contracted uterus felt beside the fetus; emergent bedside sonography.

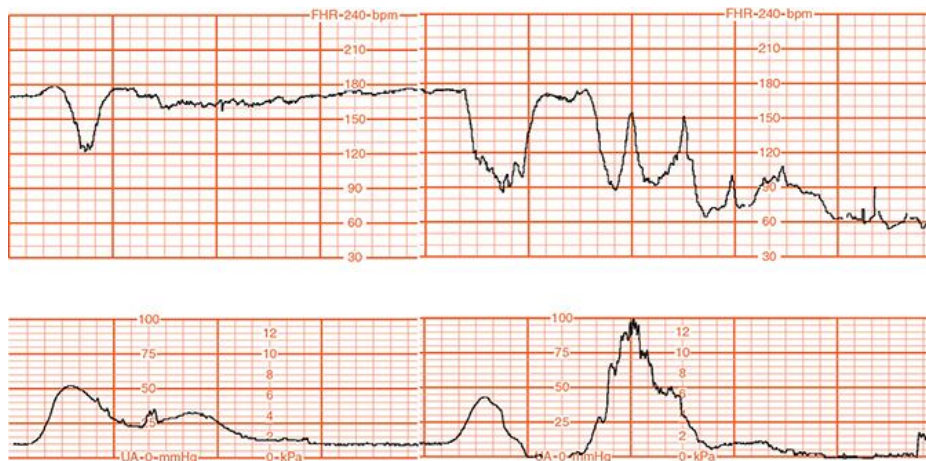


Figure 31-5. FHR tracing during rupture while pushing: a reflex push, then loss of uterine tone and worsening fetal bradycardia.

Management

- Fetal extrusion into the peritoneal cavity → poor intact survival; depends on how much placenta stays attached. **Only chance = immediate delivery, usually laparotomy**. Total placental separation → very few intact survivors.
- **Utah (35 ruptures):**
 - Decision-to-delivery **<18 min** (n = 17): **no adverse neurological outcome**.
 - >18 min (n = 18): 3 long-term impairments at **31, 40, 42 min**.
 - No deaths; severe neonatal neurological morbidity **8%**. Delivery within **30 min** did not prevent every low cord pH/Apgar, but when rupture was identified and delivery achieved **<30 min**, there was **no long-term neonatal morbidity**.
- **MFMU**: neonatal encephalopathy in **6%** (7 of 114) of TOLAC ruptures. Neonatal death after rupture **5%** (Sweden).
- **Maternal death** uncommon: **0.2%** of ruptures (Canada); **30%** in rural India.
- **Hysterectomy** may be needed; selected cases → **suture repair** with uterine preservation. **Check the bladder** (comorbid injury).

7. Multiple Repeat Cesarean Deliveries

- Complications rise with each repeat cesarean.
- **Placenta accreta spectrum (with previa): 11% at the 2nd cesarean → 67% at the 5th** (MFMU, 30,132 women).
- Almost **half of Parkland cesarean hysterectomies** are in women with ≥ 1 prior cesarean.
- Also rising: **bowel/bladder injury, ICU/ventilation, maternal mortality, operative time, hospital stay, adhesion-related bowel obstruction and pelvic pain.**
- **UKOSS (≥ 5 cesareans vs one prior):** major hemorrhage **18x**, visceral damage **17x**, critical care **15x**, delivery < 37 wk **6x**; much of this in the **18%** with previa or PAS. Neonatal complications mainly from **indicated preterm birth.**

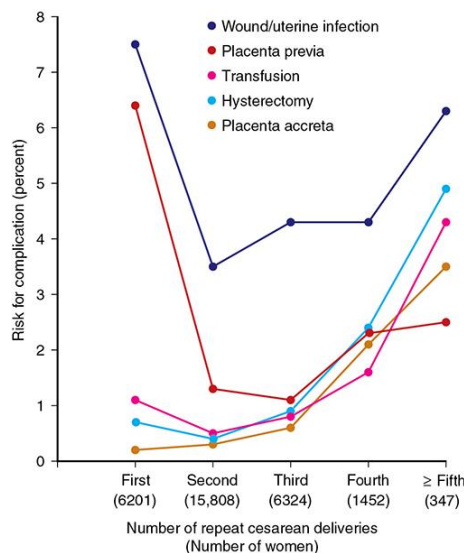


Figure 31-6. MFMU: accreta, hysterectomy and transfusion rise steeply after the third repeat cesarean.

8. Vaginal Birth After Cesarean — 2021

- **No large RCTs**; most studies compare actual, not intended, route.
- Common-sense approach: **shared decision-making** (decision aids help), with the partner if she wishes.
- Absolute adverse rates are low, but **TOLAC has higher maternal and neonatal morbidity and mortality than ERCD.**
- **Counseling includes:** documentation of the **prior incision**, risks/benefits/likely outcomes of each route, and **future-pregnancy risks.** Begin preconceptionally; keep options flexible until delivery.
- Women choosing TOLAC despite a higher-risk factor → **additions to the consent form** (ACOG).

Table 31-5. Some Recommendations of Professional Societies Concerning TOLAC

Society	Counseling	Facilities	Other
ACOG (2019d)	Offer to most women with 1 prior LTCD ; consider for 2 prior LTCDs	Safest with ability for immediate CD ; patients should be allowed to accept increased risk when not available	Not precluded: twins, macrosomia, prior low vertical or unknown incision type
SOGC	Offer to women with 1 prior LTCS; with >1 prior CD then VBAC likely successful but increased risks	Should deliver in hospital in which timely CD is available; the woman and provider must be aware of hospital resources and availability of obstetrical, anesthetic, pediatric, and operating room staff	Oxytocin or Foley catheter induction safe, but PGs not recommended; macrosomia, diabetes, postterm pregnancy, twins are not contraindications
RCOG (2015)	Offer to most women with singleton, cephalic fetus at ≥37 weeks with 1 prior LTCD ; with >1 prior CD then VBAC likely successful but increased risks, offer after counseling by senior obstetrician, VBAC checklists should be considered	Suitable delivery suite with continuous intrapartum care and monitoring ; immediate cesarean delivery and advanced neonatal resuscitation	Uncertainty about safety and efficacy with postterm, twin gestation, macrosomia, and advanced maternal age; induction and augmentation associated with two- to threefold increased risk of uterine rupture

ACOG = American College of Obstetricians and Gynecologists; CD = cesarean delivery; LTCS = low transverse cesarean delivery; PG = prostaglandins; RCOG = Royal College of Obstetricians and Gynaecologists; SOGC = Society of Obstetricians and Gynaecologists of Canada; TOLAC = trial of labor after cesarean delivery; VBAC = vaginal birth after cesarean.

Table 31-6. Conservative Approach to TOLAC**Review and follow ACOG guidelines**

Domain	Steps
Education and counseling	Preconceptionally: provide ACOG patient pamphlet. Prenatal care: early counseling and delivery planning; develop preliminary delivery plan; revisit delivery plan at least each trimester; be willing to alter decision; confirm facility capabilities
Risk assessment	Review and document operative note(s); review relative/absolute contraindications; reconsider risks as pregnancy progresses; careful consideration and counseling if: >1 prior transverse CD, unknown incision, twins, macrosomia
Labor and delivery	Review previous medical records/operative notes; cautions for induction: unfavorable cervix, high station (consider AROM; avoid prostaglandins ; respect oxytocin and know when to quit); continuous EFM ; beware of abnormal labor progress; respect EFM pattern abnormalities; be ready to abandon TOLAC ; recommend against home birth

ACOG = American College of Obstetricians and Gynecologists; AROM = artificial rupture of membranes; CD = cesarean delivery; EFM = electronic fetal monitoring; TOLAC = trial of labor following cesarean delivery.

Quick-Recall: TOLAC Candidate or Not?

Situation	TOLAC?
One prior low transverse	Yes (rupture 0.2–0.9%)
Two prior low transverse	Consider (ACOG); rupture 0.9–1.8%
Low vertical, no fundal extension	May be candidate
Unknown incision	Candidate unless classical suspected
Twins	Yes (rupture 0.7–1.1%)
Macrosomia	Not precluded (↓ VBAC; rupture 2.4% >4250 g)
Breech needing ECV	ECV not contraindicated
Classical / T-shaped / transfundal surgery	No — deliver 36 ⁰ / ₇ –37 ⁰ / ₇ wk
Prior uterine rupture	No — deliver 36 ⁰ / ₇ –37 ⁰ / ₇ wk
Prior open fetal surgery	No
Placenta previa	No
Home birth	Contraindicated
Misoprostol for ripening	Contraindicated
Epidural	Safe

Key Numbers to Memorize

- VBAC rate **3.4% (1980) → 28.3% (1996)**; TOLAC nadir **16%** (2006); TOLAC success peak **70%**, nadir **38%**; Parkland **15%** attempt, **86%** succeed
- Rupture: TOLAC **0.47%** (RR **20.7**, Guise); MFMU **0.7%** vs 0; failed TOLAC **5×** complications
- Perinatal mortality **0.13 vs 0.05%**; term HIE **46/100,000** vs 0
- One low transverse **0.2–0.9%**; classical **2–9%**; T **4–9%**; prior rupture lower **2–6%**, upper **9–32%**
- Classical incision in **53%** at 24–25 wk, **35%** at 28 wk, **<10%** at 32 wk
- Interdelivery **≤18 mo → 3×**; interpregnancy **<6 mo → 3×**
- VBAC by prior indication: **90 / 80 / 60%** (nonrecurring / fetal compromise / arrest)
- VBAC by BMI: **85 / 78 / 70 / 61%**
- Oxytocin induction rupture **1.1%** (1.8% with no prior vaginal birth) vs spontaneous **0.4%**
- LUS thickness **<2.0 mm high risk, ≥2.5 mm low risk**
- Decision-to-delivery **<18 min**: no neuro injury; neonatal encephalopathy after rupture **6%**; maternal death **0.2%**
- PAS with previa **11% → 67%** (2nd → 5th cesarean); ≥5 cesareans: hemorrhage **18×**, visceral injury **17×**
- ERCD at **≥39 wk**